

Report 2

Biomass-to-Jet: Using Gasification and Fischer–Tropsch Chemistry

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CHE 397: Senior Design II

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1. Introduction

1.1 Problem Statement

Aviation is one of the most energy-intensive transportation modes and is inherently difficult to decarbonize due to strict safety regulations, long asset lifetimes, and the absence of cost-effective alternatives to high-density liquid fuels. Conventional jet fuel is derived from fossil crude oil through distillation and hydrotreating, and its combustion contributes significantly to global greenhouse gas emissions, as well as particulate matter and NO_x, which impact both climate and air quality. Worldwide, aviation accounts for 2% of all carbon dioxide (CO₂) emissions and 12% of all CO₂ emissions from transportation. Global and national policy initiatives are establishing targets to decarbonize the aviation sector. ICAO's CORSIA program caps net CO₂ emissions from international aviation at 2020 levels through 2035, and the aviation industry has set a goal of reaching net-zero emissions by 2050. Because electrification and hydrogen are unlikely to be widely deployed in the near term, Sustainable Aviation Fuel (SAF) represents the most practical pathway to meet these objectives. In the U.S., the 2021 Sustainable Aviation Fuel Grand Challenge aims to scale domestic SAF production to 3 billion gallons annually by 2030 and 35 billion gallons annually by 2050, with a lifecycle reduction in GHG emissions of at least 50%, thereby creating long-term demand and policy support for technologies such as FT-SPK.

Sustainable Aviation Fuel (SAF) is an alternative fuel made from non-petroleum feedstocks such as biomass, waste oils, agricultural residues, and municipal solid waste. SAF has emerged as the most viable pathway for reducing aviation emissions because it is a drop-in fuel

that can meet the same performance standards as Jet A without requiring modifications to jet engines, fuel systems, or airports. This is critical for near-term decarbonization because the physical and regulatory constraints of aviation make it impractical to implement rapid changes in propulsion systems, fuel storage, and aircraft design. SAF must therefore meet stringent performance specifications, including energy density, combustion characteristics, volatility, and cold-flow properties, while offering substantial lifecycle reductions in greenhouse gas emissions.

Multiple production pathways for SAF have been developed and approved under ASTM International standards, including Hydro processed Esters and Fatty Acids (HEFA), Alcohol-to-Jet (ATJ), and Fischer-Tropsch Synthetic Paraffinic Kerosene (FT-SPK). Among these, FT-SPK has gained significant attention due to its ability to convert a diverse range of low-value, sustainable feedstocks into high-quality paraffinic hydrocarbons with ultralow sulfur and aromatic content. When produced from biomass or waste-derived syngas, FT-SPK can provide substantial lifecycle emission reductions and additionally support waste diversion efforts, contributing both environmental and societal benefits.

Despite these advantages, SAF production remains limited by high capital intensity, feedstock logistics, and operational complexity, resulting in production volumes that comprise less than 1% of global jet fuel demand. However, the combination of international emissions mandates, national policy incentives, and long-term corporate sustainability commitments is driving rapid development of the SAF sector. In this context, evaluating the technical, economic, and environmental performance of FT-SPK production is essential to determine optimal process configurations, assess scalability, and identify barriers to commercialization. This project focuses on the design and preliminary analysis of a commercial-scale FT-SPK process capable of supporting the emerging demand for low-carbon aviation fuels.

1.2 Market Analysis

The Sustainable Aviation Fuel (SAF) market is poised for significant growth, driven by increasing environmental concerns, regulatory pressures, and advancements in fuel production technologies. Despite its promising potential, SAF production faces challenges, including high capital costs for establishing production facilities, difficulties in securing consistent feedstocks, and operational complexities. These factors contribute to the high production cost of SAF, which currently ranges from \$3.50 to \$6.00 per gallon, significantly higher than traditional jet fuel, which typically costs between \$1.50 and \$2.00 per gallon. This price disparity presents a barrier to widespread adoption. However, as production scales up, technology improves, and government incentives such as tax credits, subsidies, and other financial support are implemented, the price of SAF is expected to decrease. For example, the U.S. SAF Grand Challenge aims to produce 3 billion gallons of SAF annually by 2030, which will help lower production costs through economies of scale and infrastructure development.

The global SAF market is projected to grow from USD 2.7 billion in 2025 to USD 28.6 billion by 2032, at a CAGR of 39.95%. North America is expected to remain the largest market, accounting for 46% of the market share by 2024, largely due to strong government-backed initiatives such as the SAF Grand Challenge. Europe is also a key player, driven by the European Green Deal and national goals for carbon-neutral aviation. Meanwhile, the Asia-Pacific region is set to experience rapid growth, fueled by rising investments, increasing demand for cleaner fuels, and technological advancements in SAF production. The combination of these factors, alongside the scaling of SAF production, is expected to drive down costs and make SAF more competitive with traditional jet fuel, supporting its long-term adoption in the aviation industry.

1.3 Judging Criteria

The Fischer-Tropsch (FT) pathway was selected over alternative sustainable aviation fuel production pathways, based on its superior feedstock flexibility, blending capability, scalability, long-term feasibility for meeting the growing SAF demand, and its potential for reducing carbon emissions during the life cycle of the product.

One major advantage of FT is its ability to process a broad range of carbon-based feedstocks like woody biomass, agricultural, Municipal Solid Waste (MSW), etc. The design basis of the plant accounts for this advantage by choosing St. James Parish, Louisiana, where forestry residue is abundant at a low cost. In contrast, other competing processes like HEFA are constrained almost exclusively to rich lipid feedstocks which are highly limited. These feedstock sources are heavily competed for by the renewable diesel market, leaving little to no room for the SAF market. Alcohol to Jet (ATJ) feedstocks depend on sugar, starch, or cellulosic ethanol. Food and agricultural waste have a highly competitive market as these feeds are often used in biofuel industries to produce things like biodiesel. Because FT can utilize non-food, non-lipid, waste biomass at very large scales, it provides a more robust and sustainable feedstock supply chain.

Fischer-Tropsch derived synthetic paraffinic kerosene (FT-SPK), also known as jet fuel, is certified under the ASTM D7566 Annex A1 to be blended at up to 50 vol% with conventional Jet A/A-1 jet fuel. This high blending limit provides two critical advantages. It enables significant displacement of fossil fuel derived jet fuel per unit of SAF produced and allows for drop-in use without any modifications to aircraft engines, fuel handling systems, or airport logistics that are designed to operate with fossil fuel derived kerosene. Other pathways offer a lower maximum blend limit ranging from 10% to 30% thus FT offers both high blending capabilities and feedstock scalability.

The FT pathway also enables deep decarbonization because the biomass to liquids routes achieves significant carbon intensity reductions depending on feedstock logistics and hydrogen sourcing. Other pathways simply do not offer this level of decarbonization due to FT's ability to reduce greenhouse gas emissions.

The long-term expansion potential for FT-SPK is very promising as SAF initiatives and demands continue to increase. Existing commercial plants demonstrate that 10,000 barrels/day is sustainable and efficient making the design of this plant strategic. Competing processes are generally smaller in scale, are limited by supply of feedstocks, and are less aligned with long-term SAF volume requirements mandated by U.S and EU policy. The U.S. Gulf Coast also already supports biomass supply chains, gasification and syngas-handling experience, hydrogen production and contains existing refinery and upgrading systems.

In summary, the Fisher-Tropsch pathway was selected because it provides broad and abundant feedstock options, 50% blend certification, which matches or exceeds alternatives, superior scalability for large volume SAF production, and lower carbon intensity. For these reasons, FT to SAF offers the most robust, sustainable, and scalable pathway for meeting long-term aviation decarbonization targets compared to methods like HEFA or ATJ.

2. Process Description

2.1 Chosen Process

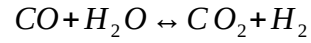
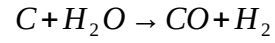
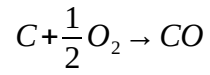
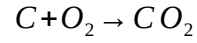
The general Fischer-Tropsch process to produce synthetic paraffinic kerosene, also known as Jet Fuel, is as follows. The first step is feedstock selection, which can be any carbon-based material known as biomass. In this case, wood-based biomass is gasified to create synthetic gas, syngas, which is primarily a mixture of carbon monoxide and hydrogen gas ($\text{CO}_{(g)}$)

+ $\text{H}_2(\text{g})$). The syngas must then undergo cleaning to ensure that a relatively low concentration of carbon dioxide (CO_2) remains. The lean syngas then undergoes catalytic reactions in an FT reactor that uses cobalt as a catalyst in low temperatures to form long chain hydrocarbons. These hydrocarbons are then sent to upgrading and fractionation processes to produce synthetic paraffinic kerosene (SPK) that meets the ASTM D7566 standard. The final product once blended at up to 50% must meet the new ASTM D1655 standard.

The detailed process will follow the major sections of the block flow diagram highlighted below in (citation) which include biomass gasification and syngas cleanup, Fischer-Tropsch reactions, hydrotreating, and hydrocracking. The operating conditions are based on the process simulation modeled using ASPEN Plus v14

The process begins with the gasification of wood-based biomass to produce syngas which is then cleaned to produce lean syngas that are fed into the FT reactor. For this process, the necessary feed is found in the design basis section to be approximately 5,700 tonnes/day of biomass. Unlike fossil-based syngas production, biomass gasification produces syngas containing significant amounts of CO_2 , particulates and ash, as well as other contaminants. Extensive gas cleanup and conditioning are required prior to FT synthesis.

The first step in the biomass gasification process, shown in Figure 1, is to heat and dry the incoming biomass in a dryer (*D-100*) operating at about 250°F to reduce the typical inlet moisture of 30-50 wt.% down to <10 wt.%. Air is fed into the dryer after being heated and it expels the dry biomass and excess moisture. The feed is then ready to be sent to the biomass gasifier (*R-100*) where the dried biomass along with oxygen is converted into raw syngas via partial oxidation reactions with oxygen. The primary reactions are shown below (Equation 1).



Equation 1: Primary Biomass Conversion Equations

Assuming a fluidized bed reactor is used then the temperature can be specified at ~1400°F with pressure an operating pressure of ~15 psia. The products of this reactor include raw syngas and tar.

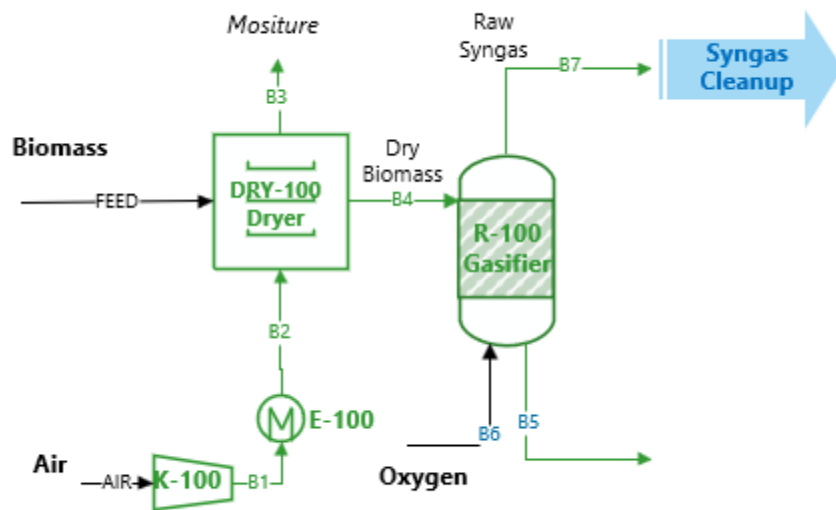
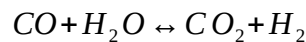


Figure 1: Biomass Gasification Process Flow Diagram

The cleanup process is as show in *Figure 2* below. Raw Syngas is fed into the cyclone *C-100* removes fine ash and char particles from raw syngas to protect the downstream equipment and operates at ~1400°F. The incoming gas will be compressed to a new pressure of 180 psia. E-100 then cools the hot syngas to enable downstream wet scrubbing and contaminant removal. Following cooling, the syngas scrubber *V-100* removes remaining contaminants using water and operates at 100°F and ~180psia. The knockout drum *V-101* removes entrained liquid droplet and condensed liquids from scrubbed syngas at 86°F. The water-gas shift reactor *R-101* adjusts the H₂/CO ratio to approximately 1.7, as determined in the design basis and required for cobalt FT synthesis. The main reaction here is shown below (Equation 2). The reactor is expected to operate at high temperatures ~212°F) and at approximately atmospheric pressure 14.5 psia. From the design basis the steam to dry syngas ratio can be taken to be 0.5kg/kg.



Equation 2: Water-Gas Shift Reaction

The final *V-102* absorber reduces the concentration of CO₂ to improve FT reactor productivity and operates at 145 psia removing 70-90% of CO₂. The resulting lean, conditioned syngas meets the FT feed specifications of having a H₂/CO ratio of 1.7, particulate concentration < 0.1 mg/Nm³, and negligible concentration of sulfur and nitrogen species.

Key assumptions include that woody residues will have a consistent composition, air-blown gasification is acceptable, heat integration will greatly reduce external utility demands, and overall biomass to syngas efficiency aligns with reported ~70% cold gas efficiency.

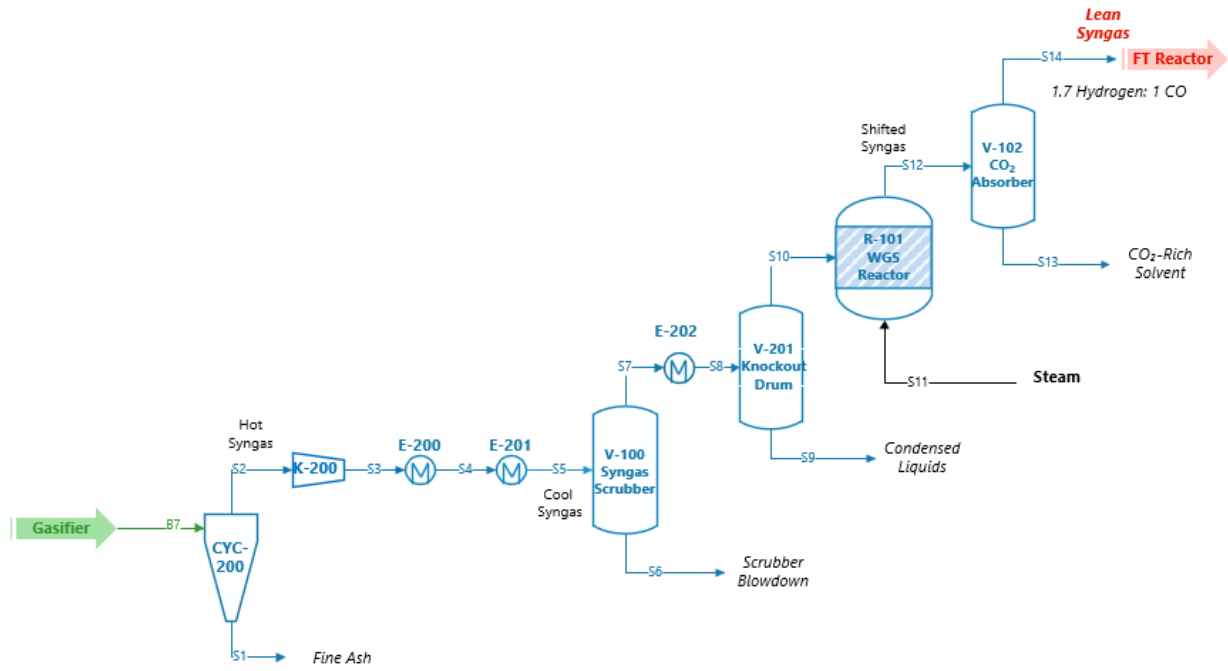
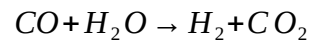
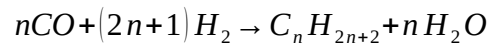


Figure 2: Syngas Cleanup Process Flow Diagram

Following the production of lean syngas, this is fed into the reactor section of the process *Figure 3*. This part of the process converts the conditioned syngas into long chain hydrocarbons producing waxes and middle distillates via the following chemical equations (Equation 3).



Equation 3: Fischer-Tropsch Reactions

The design basis will cover the optimal CO/H₂ ratio in the syngas feed. As previously mentioned, the FT-synthesis will occur in a fixed bed reactor at ~600°F and 290 psia with a 20

wt.% cobalt supported catalyst. The FT reactor produces hydrocarbons with a molecular weight distribution which can be described by the Anderson-Schulz-Flory distribution (Equation 4) .

$$\frac{W_n}{n} = 1 - \alpha^2 \alpha^{n-1}$$

Equation 4: Anderson-Schulz-Flory distribution

The catalyst was chosen due to Cobalt's high activity for long-chain paraffins in the desired SAF range of C₈-C₁₆. Alternatives not chosen include iron because it is less selective towards middle distillates, and nickel because of concerns of excessive methane formation. Key assumptions include those of the design basis.

High-pressure (HP) V-300 and low-pressure (LP) V-301 3-phase separators are added to separate heavy waxes (C₂₀+) from liquid hydrocarbons and further separate liquid hydrocarbons into naphtha, middle distillates, and light hydrocarbons operating at ~300 psia and ~200 psia, respectively. Wax from the HP drum is sent to hydrocracking, and the fuel from the LP drum is vented. The LP drum V-301 separates water and light hydrocarbons, removes inert gases, and vents flue gas. The light hydrocarbons are sent to the hydrocracking unit, and the upgrading processes follow.

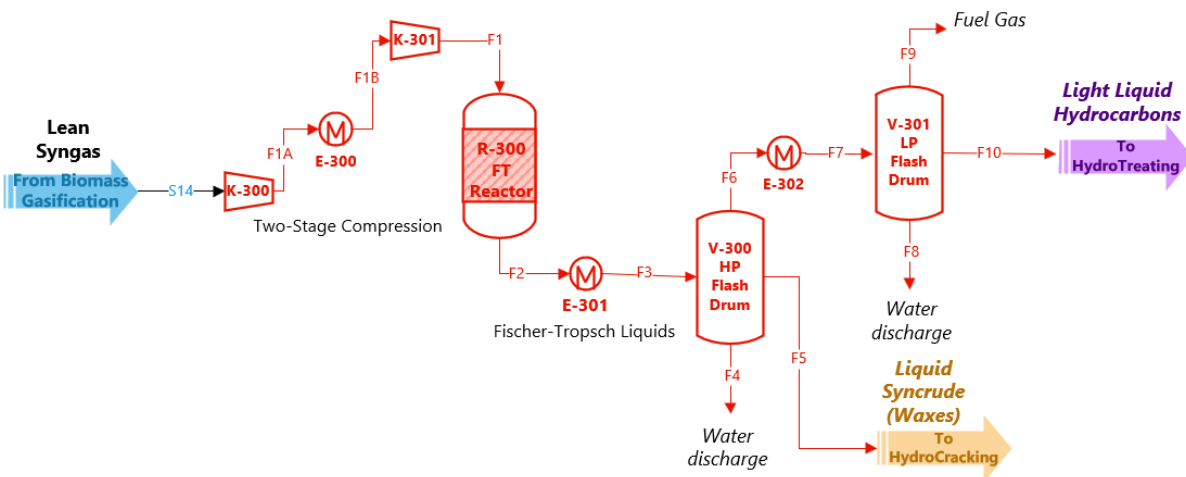


Figure 3: Fisher-Tropsch Synthesis Process Flow Diagram

The cobalt-catalyzed Fisher-Tropsch liquids which include linear paraffins, with a large fraction of heavy waxes (C_{20}^{+}) and a smaller fraction of middle distillates (C_9-C_{20}) must be sent to upgrading processes like hydrotreating and hydrocracking (Figure 4-Figure 5). While the hydrocarbons produced are sulfur and aromatics-free, they do not meet aviation fuel specifications due to excessive molecular weight and lack of isomerization. Hydrocracking and hydrotreating are required to crack FT waxes into jet fuel carbon range, isomerize linear paraffins into branched iso-paraffins to meet SAF freezing point limits, and produce co-product naphtha.

In the hydrotreating (Figure 4) section R-400 serves as the primary hydrotreating reactor for FT liquids ensuring hydrogenation of olefins and removal of trace oxygenates. The feed includes FT light liquid hydrocarbons, and a blue hydrogen recycle stream. Unreacted hydrogen is recycled and supplied with purchased makeup hydrogen to comprise the hydrogen feed into

the reactor. The feed is pumped and heated prior to entering *R-400*. Typical operating conditions include temperatures at ~450 °F and 900 psia. The flash drum *V-400* supplies unreacted hydrogen to the recycle stream and hydrotreated hydrocarbons into the first distillation column of the section. The primary fractionation column *C-400* separates light ends, naphtha, and unconverted waxes + jet fuel bottoms via atmospheric distillation. The column will typically operate at a top temperature of 150°C with a bottom temperature at 330-360°C at 1-2 bar. The final fractionation occurs in the finishing vacuum column *C-401* to separate SAF from heavier diesel-range hydrocarbons. Operating conditions here are like refinery kerosene fractionators and main products include SAF and diesel co-product.

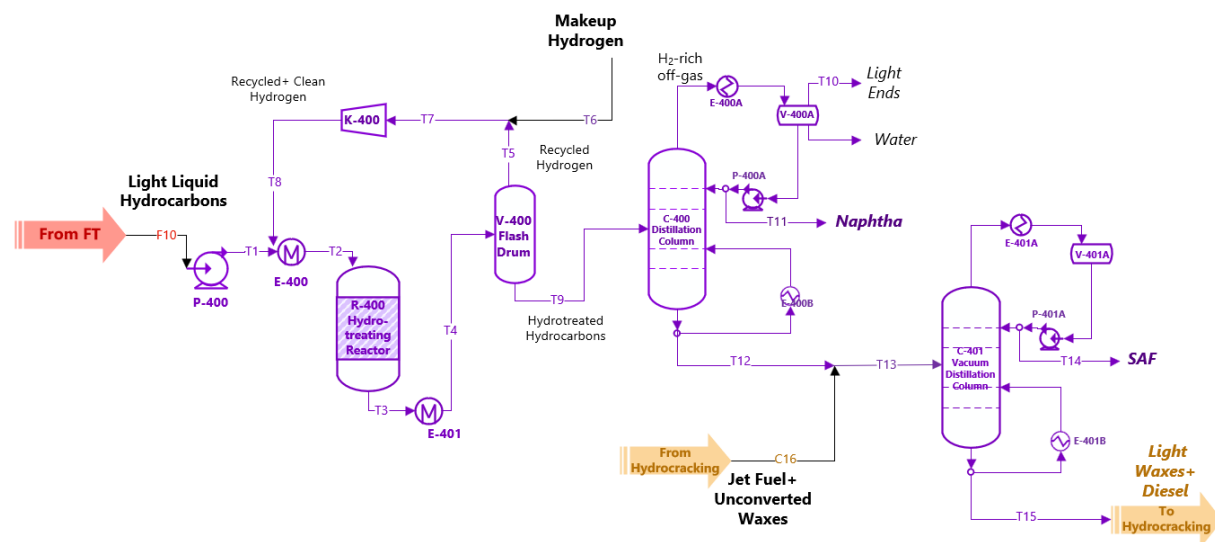


Figure 4: Hydrotreating Process Flow Diagram

The hydrocracking section (*Figure 5*) serves to crack FT wax into lighter paraffins while simultaneously isomerizing products. Feed streams include light waxes from hydrotreating (*T16*), liquid Syncrude from the FT synthesis, and makeup blue hydrogen (*C10*). A similar hydrogen gas recycle stream from the hydrotreating section is adopted here. The hydrocracker

typically operates at 350°F and 900 psia. The conditions are chosen to promote isomerization and high wax conversion. The V-500 3-phase separating drum separates unreacted hydrogen, water discharge, and liquid cracked hydrocarbons at 120°F and 300 psia. The fractionation column *C-500* recovers Naphtha and light ends. Unconverted waxy material is recycled back to the hydrotreater to maximize SAF yield and overall carbon efficiency.

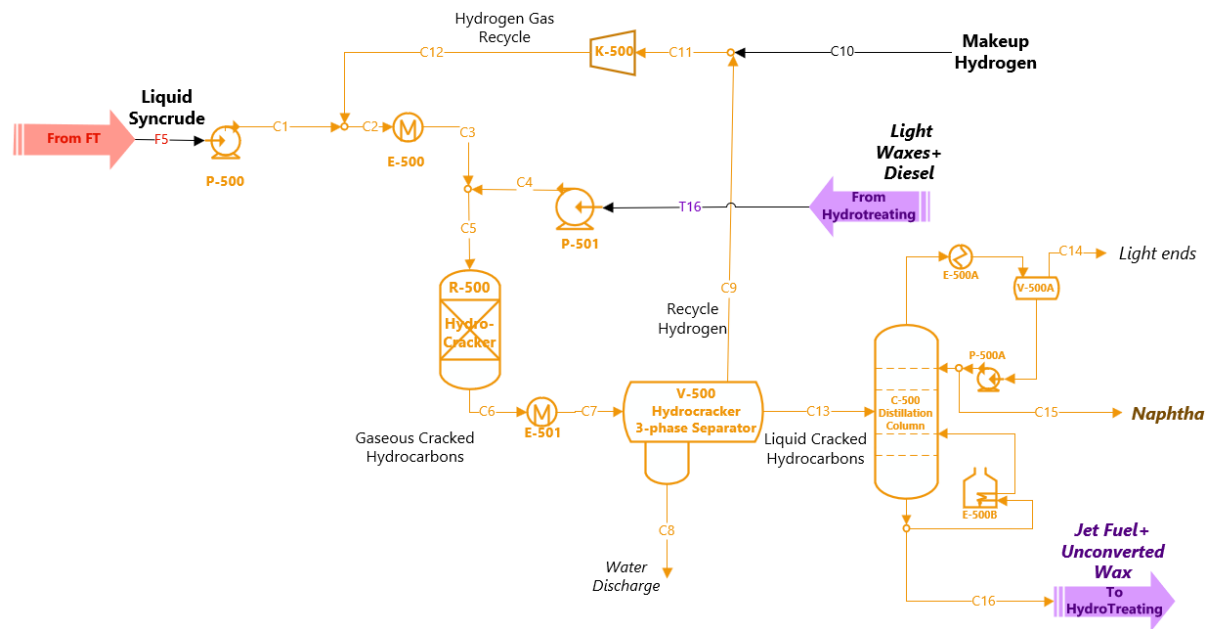


Figure 5: Hydrocracking Process Flow Diagram

Key assumptions in the upgrading section include that FT liquids contain negligible sulfur and nitrogen, eliminating the need for deep hydrotreating. The hydrogen is also assumed to be externally supplied from regional U.S Gulf Coast infrastructure. The hydrocracking severity is tuned to maximizing jet fuel range rather than gasoline.

2.2 Competing Processes

One of the main competing processes of Fischer-Tropsch process is the Hydro processed Esters and Fatty Acids (HEFA/HVO) process. Like the name suggests, hydro processed esters and fatty acids are the feed stock for this process with hydrotreated vegetable oil as another option. The first step of the process is pretreatment to remove any impurities. Pretreatment is crucial for SAFs especially since unlike traditional crude oil, it contains contaminants like phosphorus and metals. The pretreatment follows 5 steps: feedstock blending, straining, caustic treatment/water washing, impurity adsorption, and vacuum and filtration. Post pre-treatment hydrogenation and hydrodeoxygenation occur. Hydrogenation saturates any double bonds in the fatty acid chains, stabilizing them. Hydrodeoxygenation is where oxygen is removed via the creation of water (H_2O). Both processes are catalyzed, the first is typically by Pd/C, Pt/C, Ru/C or Raney Ni for hydrogenation and Ni, Co, Mo, W and Al_2O_3 as typical catalysts for hydrodeoxygenation.

Following Hydrotreating, the product contains a large concentration of high carbon range materials C15- C18 paraffins, and this material is typically hydrocracked down to a useful range of carbons for SAFs (C8-C16). After the material is cracked there is a large concentration of straight chains that must be converted into branched isomers. The branched isomers have differing physical properties that are desirable for aviation fuel. These isomers control the fuel's ignition time, freezing point, viscosity, density, and flash point all of which must fall within aviation limits. After the isomers are created, they are separated via a high-pressure hot separator, cold high-pressure separator, and then fractionator columns. This leaves the products jet-fuel, diesel or naphtha. After this, any remaining gas product has hydrogen recovered and is then recycled back into the hydrogenation reactor.

The HEFA process is not without its limitations and shortcomings. Primarily it is more economical for these plants to develop renewable diesel over SAFs. Secondly, feedstocks have existing uses such as food production. And finally, it is more rigid in its production of carbon ranges, typically producing only C8-C16. FT synthesis is able to be tailored to a desired range of carbons for various uses. The block flow diagram for a HEFA process is shown below (Figure 6).

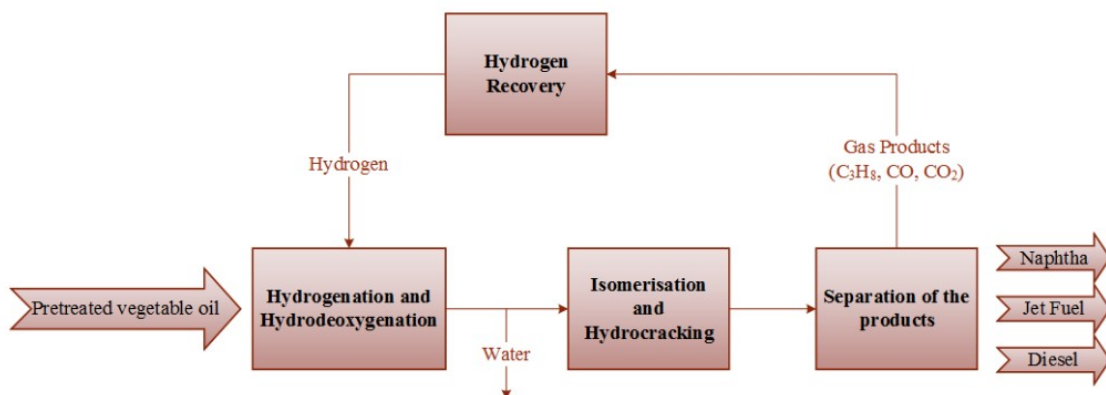


Figure 6: HEFA Block Flow Diagram

Another major competing process for Fischer-Tropsch synthesis is the alcohol to jet fuel (ATJ) route. The first step in the ethanol to jet process is the catalytic fueled dehydration of ethanol to ethylene. Typical catalysts for this process include HZSM-5 which is a zeolite and rare earth metals such as Lanthanum. HZSM-5 is the most common catalyst but is highly acidic, leaving it susceptible to coking. This results in a lower lifetime and large catalyst cost and regeneration rate. As a result, the ATJ process has a large catalyst cost and leaves room for improvements. Following the dehydration, the product is oligomerized. The olefins are then combined into larger carbon chains typically from the C8-16 range. Typical catalysts for this step include zeolites and Ni-based. The typical choice for catalysts is Ni-ALSBA-15 due to a large conversion of ethylene at 95% and a high selectivity for the C5-C7 range.

Following the oligomerization, the material is often sometimes separated before it is hydrogenated, but this step is not necessarily done in this way. The last step of the process is hydrogenation, and this process is typically done through two separate reactors with two different catalysts. The first reactor typically targets the shorter hydrocarbons in the C8-C11 range while the second reactor achieves a higher conversion of the larger hydrocarbon chains. Typical catalysts for the hydrogenation step include Ni-C and Pt/AL₂O₃. Following hydrogenation, the product is separated via distillation.

Similarly, HEFAs/HVOS alcohol to jet fuel has its limitations. It also is limited in its carbon ranges of C8-C16. It does offer a more flexible feedstock than HEFA but lacks a large commercial scale. It is limited by alcohol availability in each area as alcohol is a high demand product for other industries. Overall FT synthesis offers a large amount of flexibility in product creation and a large amount of feedstock options. For these reasons the decision to go with an FT-based production pathway was made. The block flow diagram for ATJ process is shown below (Figure 7).

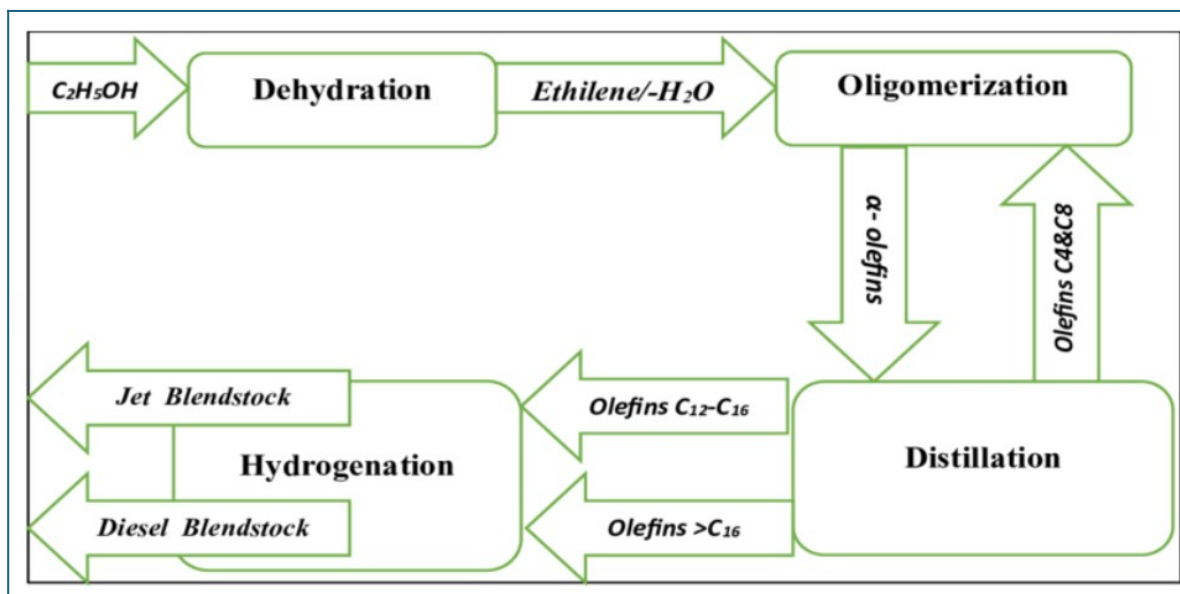


Figure 7: ATJ Block Flow Diagram

3. Appendix

Appendix I: Design Basis

This design is based on a ~10,000 barrels per day (bbl/day) sustainable aviation fuel (SAF) production facility using biomass gasification, syngas conditioning, low-temperature cobalt Fischer–Tropsch (FT) synthesis, and downstream hydrocracking/hydrotreating. The overall configuration and scale are benchmarked against commercial FT-SAF projects under development along the U.S. Gulf Coast—notably the DG Fuels plant in Louisiana, which will employ Johnson Matthey and bp’s FT CANS™ technology at a capacity of 13,000 bbl/day. This provides a realistic industrial reference point for sizing, performance expectations, and catalyst inventory at the scale considered in this study.

St. James Parish, Louisiana is selected as the plant location due to its favorable combination of biomass supply, industrial infrastructure, hydrogen availability, and proximity to emerging FT-SAF developments. Louisiana ranks among the top U.S. producers of forestry residues, logging waste, and wood-processing by-products, which form the primary biomass feedstock. The Mississippi River petrochemical corridor offers extensive utilities, existing pipelines, and refinery-adjacent infrastructure that reduce capital and operating costs for a biomass-to-liquids (BtL) facility. The region also benefits from strong hydrogen production capacity supported by ammonia and petrochemical operations, as well as growing SAF demand and supportive federal incentives such as the Inflation Reduction Act’s SAF tax credits. Deep-water access through the ports of Baton Rouge and New Orleans simplifies logistics for catalyst delivery, equipment transport, and export of finished SAF.

To meet the target of 10,000 bbl/day SAF, the FT synthesis and upgrading section is designed for approximately 1,450 tonnes/day of total FT liquids, of which about 1,250 tonnes/day fall within the SAF carbon range (C_8 – C_{16}), with the remainder as naphtha co-products. This production scale is consistent with biomass-to-liquids studies reporting overall biomass-to-liquids efficiencies of ~40% (LHV basis) for gasification followed by FT synthesis and upgrading.

At the front end of the process, the plant processes 5,700 tonnes/day of woody biomass, sourced from regional forestry residues and mill by-products. The biomass is dried on-site and fed to the gasifier to produce a raw syngas containing CO, H_2 , CO_2 , steam, and light hydrocarbons.

Biomass gasification naturally produces syngas with a low H_2 /CO ratio, which is not suitable for low-temperature cobalt FT catalysts and FT synthesis in general. Because cobalt FT operation requires a target H_2 /CO ratio near 2, the design incorporates a water–gas shift (WGS) reactor to partially convert CO and steam to H_2 and CO_2 . Typical design studies report steam-to-syngas ratios between 0.4 and 0.7 kg steam per kg of dry syngas; here a value of 0.5 kg steam per kg of conditioned syngas is selected, resulting in a steam utility load of approximately 1,000 tonnes/day for syngas conditioning at the required FT reactor feed rate.

The FT synthesis section uses a low-temperature cobalt catalyst, selected for its high activity and strong selectivity toward long-chain linear paraffins, producing heavy wax (C_{20}^+) and middle distillate (C_9 – C_{20}) hydrocarbons. Catalyst inventory is scaled from published industrial data indicating that a 100,000 bbl/day FT facility requires roughly 646 tonnes of cobalt. Normalizing to the capacity of this design yields an estimated 65 tonnes of cobalt,

corresponding to approximately 320 tonnes of supported catalyst assuming a 20 wt% Co formulation.

Downstream of FT synthesis, the heavy wax and middle-distillate fractions are routed to hydrocracking and hydrotreating units to produce the desired blend of SAF and naphtha. The detailed hydrogen consumption associated with these upgrading steps is addressed separately in the mass balance section of the report.

Appendix II: Block Flow Diagram

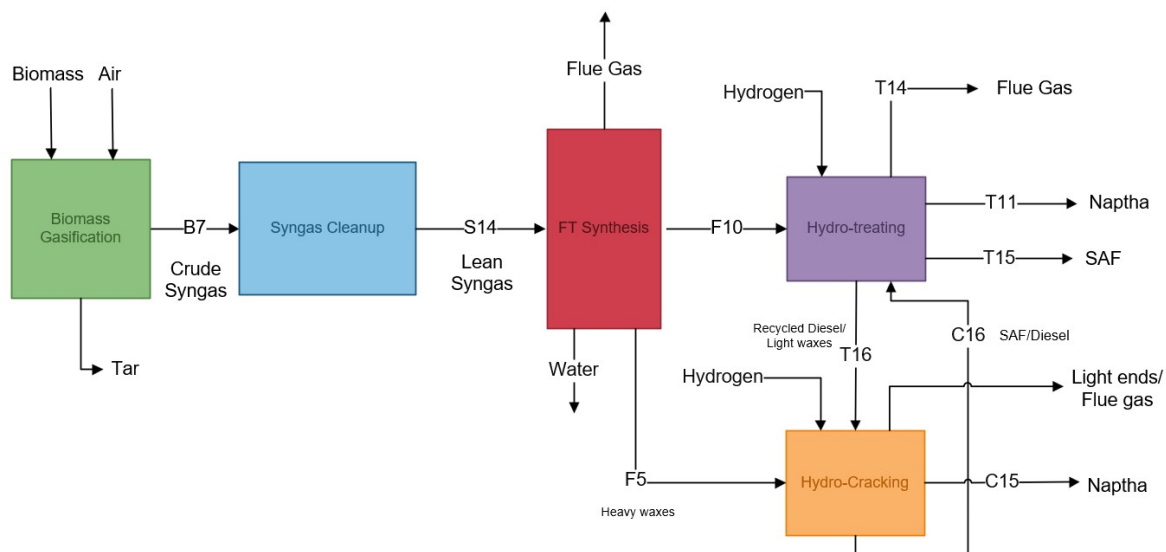


Figure 8: Final Block Flow Diagram

Appendix III: Process Flow Diagram

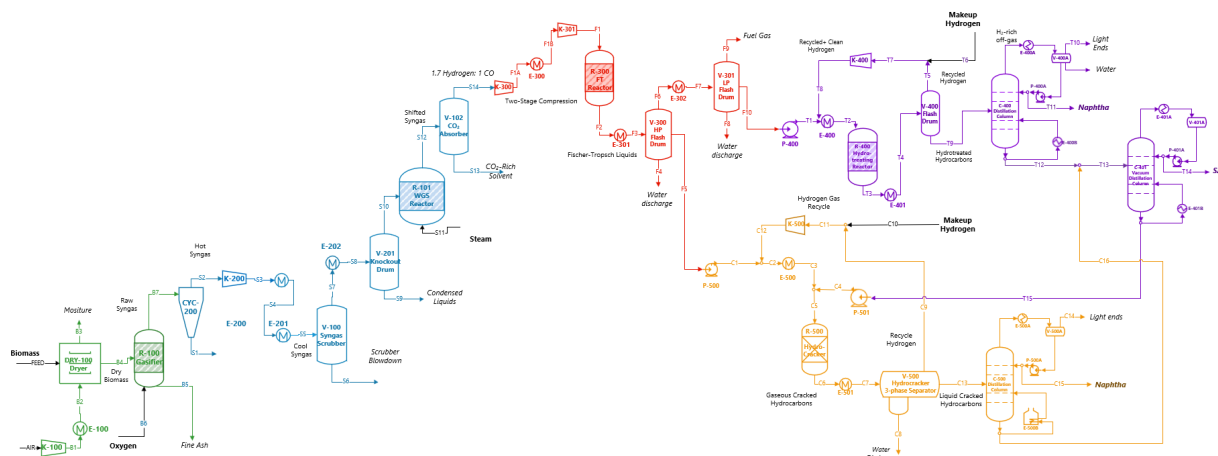


Figure 9: Final Process Flow Diagram

Appendix IV: Material Balance

The overall material balance is based on the final Aspen Plus v14 simulation and is organized by the major blocks in the flowsheet: Biomass Gasification (100), Syngas Cleanup/Conditioning (200), Fischer–Tropsch Synthesis (300), Hydrotreating (400), and Hydrocracking (500). The final configuration reflects the process changes made during optimization, most importantly the reduced biomass basis and the decision to eliminate diesel as an external co-product by routing diesel-range material and heavy wax back into the upgrading loop, so that they are ultimately converted into jet-range material.

The front-end biomass feed enters the process at approximately 523,598 lb/hr of dry woody biomass. After drying and the introduction of the oxidant, the gasifier produces a raw syngas stream B7 with a total flow rate of approximately 335,750 lb/hr. This stream contains approximately 261,959 lb/hr CO, 23,625 lb/hr H₂, 17,473 lb/hr CO₂, and 20,251 lb/hr CH₄, along

with water vapor, minor light hydrocarbons, nitrogen, and approximately 5,044 lb/hr of ash. The cyclone removes the solid ash as stream S1, reducing the vapor-phase syngas flow to approximately 330,706 lb/hr in stream S2. No significant hydrocarbon loss occurs in this solids separation step, and carbon is conserved within the gas phase.

The syngas cleanup section conditions the gas for catalytic synthesis. After cooling, scrubbing, and condensation, water is partially removed in liquid knockout streams (S6 and S8), reducing the water loading prior to shift conversion. The water–gas shift reactor increases hydrogen concentration by converting CO, and the downstream CO₂ removal unit separates CO₂ into stream S13. Following these steps, the lean syngas stream S14 has a total mass flow of approximately 245,086 lb/hr. This stream contains approximately 218,321 lb/hr CO and 26,765 lb/hr H₂, with CO₂ reduced to negligible levels and no remaining solids. The reduction in total mass from the raw syngas basis reflects the removal of CO₂, condensed water, and solids. The S14 stream represents the total fresh syngas feed to the FT reactor before recycle blending.

In the Fischer–Tropsch section, the reactor feed F1 equals the conditioned syngas flow of approximately 245,086 lb/hr. Across the reactor, total mass is conserved; however, the species distribution shifts substantially. The reactor effluent F2/F3 includes approximately 33,111 lb/hr of reaction water formed, 34,118 lb/hr CO₂ generated or carried through, and multiple hydrocarbon fractions. Hydrocarbon products are represented in lumped carbon-number ranges: C₅–C₇ (~14,802 lb/hr), C₈–C₁₆ (~46,932 lb/hr), C₁₇–C₂₂ (~38,526 lb/hr), and C₂₃–C₄₀ (~23,975 lb/hr), with trace C₄₂–C₆₀ components. High- and low-pressure flash separation splits this effluent into a hydrocarbon liquid stream, F5 (approximately 110,182 lb/hr), and a gas stream, F6 (approximately 125,787 lb/hr). The gas stream contains unreacted CO and H₂, light hydrocarbons, CO₂, and water vapor and is partially recycled to maintain reactor utilization, with

a controlled purge to prevent inert buildup. The liquid stream F5 contains the majority of the C8+ material and serves as the feed to the upgrading unit.

Hydrotreating receives the FT liquid stream F10, which represents the stabilized portion of F5 after initial separation, with a flow rate of approximately 18,754 lb/hr, as shown in the detailed stream table. Hydrogen addition and mild conversion redistribute minor quantities of light gases and water but do not significantly alter total hydrocarbon mass. Across the hydrotreating reactor and flash separation sequence (T1–T15), total hydrocarbon mass is conserved within expected hydrogen addition and purge limits. The stabilized liquid leaving the hydrotreating block is then fractionated to separate lighter naphtha-range components from heavier material directed to hydrocracking.

The hydrocracking section processes the heavy wax fraction originating from the FT separation (C23–C40 and C17–C22 material) along with heavier hydrotreated streams. The wax feed case represented by stream F6 entering the hydrocracking block shows approximately 26,222 lb/hr of heavy hydrocarbons prior to hydrogen addition. After mixing with hydrogen (C2/C3), the total reactor feed increases to approximately 51,095–52,430 lb/hr due to hydrogen incorporation. Within the hydrocracker, C20+ components are cracked and isomerized into lighter fractions, redistributing mass into C5–C7, C8–C16, and C17–C22 ranges while generating a light gas fraction. Reactor outlet separation removes a vapor-phase stream containing C1–C4 and light components, while the liquid phase proceeds to fractionation. Unconverted heavy material is internally recycled within the hydrocracking loop to increase overall conversion of wax into jet-range hydrocarbons.

When hydrotreated and hydrocracked liquid streams are combined, the total liquid production aligns with approximately 1,450 tonnes/day of finished fuels. Of this total,

approximately 1,250 tonnes/day are recovered in the C8–C16 SAF range, with the balance primarily in the naphtha range. Diesel-range hydrocarbons are not withdrawn as a product stream in the final configuration but are internally recycled to the hydrocracking section. Overall mass closure across the entire flowsheet is maintained within Aspen's numerical tolerance, and the stream data confirm conservation of total mass through each major block, with expected redistribution among the vapor, liquid, and solid phases.

Table 1: Mass Balance Biomass Gasification Section-100

Component	Units	Feed	Air	B1	B2	B3	B4	B5	B6	B7
Phase	Unitless	Solid	Vapor	Vapor	Vapor	Vapor	Solid	Solid	Vapor	Vapor
Temperature	0F	77	77	309	140	250	250	1472	77	1,397
Pressure	psia	14.7	19.7	48.7	43.7	14.7	14.7	15	15	15
Biomass	lb/hr	523,598	0	0	0	0	327,249	0	0	
CO	lb/hr	0	0	0	0	0	0	0	0	261,959
H ₂	lb/hr	0	0	0	0	0	0	0	0	23,625
H ₂ O	lb/hr	0	0	0	0	196,349	0	0	0	7,003
CO ₂	lb/hr	0	0	0	0	0	0	0	0	17,473
CH ₄	lb/hr	0	0		0	0	0	0	0	20,251
C ₂ H ₆	lb/hr	0	0	0	0	0	0	0	0	1
C ₃ H ₈	lb/hr	0	0	0	0	0	0	0	0	trace
O ₂	lb/hr	0	0	0	0	0	0	0	11,023	trace
N ₂	lb/hr	0	0	0	0	0	0	0	0	393
Ash	lb/hr	0	0	0	0		0	0	0	5,044
Air	lb/hr		220	220	220	220	0	0	0	0
Tar	lb/hr	0	0	0	0	0	0	2,522	0	0
Total	lb/hr	523,598	220	220	220	196,570	327,249	2,522	11,023	335,750

Table 2: Mass Balance Syngas Section-200 Part 1

Component	Units	B7	S1	S2	S3	S4	S5	S6
Phase	Unitless	Vapor	Solid	Vapor	Vapor	Vapor	Vap-Liq	Liquid
Temperature	°F	1,397	1,397	1,397	3,166	500	100	100
Pressure	psia	15	15	15	185	180	175	180
CO	lb/hr	261,959	0	261,959	261,959	261,959	261,959	trace
H ₂	lb/hr	23,625	0	23,625	23,625	23,625	23,625	trace
H ₂ O	lb/hr	7,003	0	7,003	7,003	7,003	7,003	4,958
CO ₂	lb/hr	17,473	0	17,473	17,473	17,473	17,473	trace
CH ₄	lb/hr	20,251	0	20,251	20,251	20,251	20,251	trace
C ₂ H ₆	lb/hr	1	0	1	1	1	1	trace
C ₃ H ₈	lb/hr	trace	0	trace	trace	trace	trace	trace
O ₂	lb/hr	trace	trace	trace	trace	trace	trace	trace
N ₂	lb/hr	393	0	393	393	393	393	0
Ash	lb/hr	5,044	5,044	0	0	0	0	0
Total	lb/hr	335,750	5,044	330,706	330,706	330,706	330,706	4,958

Table 3: Mass Balance Syngas Section-200 Part 2

Component	Units	S7	S8	S9	S10	S11	S12	S13	S14
Phase	unitless	Vapor	Vapor	Liquid	Vapor	Vapor	Vapor	Vapor	Vapor
Temperature	°F	100	86	86	86	212	650	650	650
Pressure	psia	180	175	175	175	15	145	145	145
CO	lb/hr	261,959	261,959	trace	261,959	0	218,321	0	218,321
H ₂	lb/hr	23,625	23,625	trace	23,625	0	26,765	0	26,765
H ₂ O	lb/hr	2,045	2,045	881	1,164	38,931	12,029	12,029	0
CO ₂	lb/hr	17,473	17,473	trace	17,473	0	86,038	86,038	0
CH ₄	lb/hr	20,251	20,251	trace	20,251	0	20,251	20,251	0
C ₂ H ₆	lb/hr	1	1	trace	1	0	1	1	0
C ₃ H ₈	lb/hr	trace	trace	trace	trace	0	trace	trace	0

O ₂	lb/hr	trace	trace	trace	trace	trace	trace	trace	0
N ₂	lb/hr	393	393	trace	393	0	393	393	0
Ash	lb/hr	0	0	0	0	0	0	0	0
Total	lb/hr	325,747	325,747	881	324,866	38,931	363,797	118,711	245,086

Table 4: Mass Balance FT Section-300 Part 1

Component	Units	S13	F1	F2	F3	F4
Phase	Unitless	Vapor	Vapor	Vap-Liq	Vap-Liq	Liquid
Temperature	°F	650	716	608	257	296
Pressure	psia	145	289	290	285	203
H ₂	lb/hr	26,765	26,765	17	17	trace
CO	lb/hr	218,321	218,321	21,713	21,713	trace
H ₂ O	lb/hr	0	0	33,111	33,111	9,113
CO ₂	lb/hr	0	0	34,118	34,118	4
H ₂ S	lb/hr	0	0	trace	trace	trace
C ₁ -C ₄ (H ⁺)	lb/hr	0	0	31,879	31,879	1
C ₁ -C ₄	lb/hr	0	0	trace	trace	trace
C ₅ -C ₇	lb/hr	0	0	14,802	14,802	trace
C ₈ -C ₁₆	lb/hr	0	0	46,932	46,932	trace
C ₁₇ -C ₂₂	lb/hr	0	0	38,526	38,526	trace
C ₂₃ -C ₄₀	lb/hr	0	0	23,975	23,975	trace
C ₄₂ -C ₆₀	lb/hr	0	0	13	13	trace
Total	lb/hr	245,086	245,086	245,086	245,086	9,117

Table 5: Mass Balance FT Section-300 Part 2

Component	Units	F5	F6	F7	F8	F9	F10
Phase	Unitless	Liquid	Vapor	Vap-Liq	Liquid	Vapor	Liquid
Temperature	°F	296	296	80	122	122	122
Pressure	psia	203	203	198	300	300	300
H ₂	lb/hr	trace	17	17	trace	17	trace
CO	lb/hr	115	21,598	21,598	trace	21,511	87

		H ₂ O	lb/hr	2,030	21,967	21,967	21,567	279	122		
		CO ₂	lb/hr	591	33,524	33,524	6	32,751	767		
		H ₂ S	lb/hr	trace	trace	trace	trace	trace	trace		
Table 6: Mass Balance Hydrotreating Section-400 Part 1											
Component	C ₁ -C ₄ (H ⁺)	lb/hr	1,354	30,525	30,525	trace	27,577	2,948		T7	T8
Phase	Unitless	Liquid	Liquid	Vapor	Vapor	Liquid	Vapor	Vapor	Vapor	Vapor	Vapor
Temperature	°F	122	435	572	608	122	122	77	101	445	
Pressure	psia	300	354	45.0	45.2	45.2	45.2	1.1	1.3	900	
H ₂	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace	trace
CO	lb/hr	87	87	162	162	75	0	75	75	75	
H ₂ O	lb/hr	122	122	123	123	1	0	1	1	1	
CO ₂	lb/hr	767	767	882	882	115	0	115	115	115	
Total	lb/hr	18,754	18,754	18,754	19,054	19,054	300	0	300	300	
H ₂ S	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace	trace
C ₁ -C ₄ (H ⁺)	lb/hr	2,948	2,948	2,948	3,045	3,045	97	0	97	97	
C ₁ -C ₄	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace	trace
C ₅ -C ₇	lb/hr	7,589	7,589	7,589	7,600	7,600	11	0	11	11	
C ₈ -C ₁₆	lb/hr	7,119	7,119	7,119	7,120	7,120	1	0	1	1	
C ₁₇ -C ₂₂	lb/hr	116	116	116	116	116	trace	0	trace	trace	trace
C ₂₃ -C ₄₄	lb/hr	7	7	7	7	7	trace	0	trace	trace	trace
Total	lb/hr	18,754	18,754	18,754	19,054	19,054	300	0	300	300	

Table 7: Mass Balance Hydrotreating Section-400 Part 2										
Component	Units	T9	T10	T11	T12	C16	T13	T14	T15	
Phase	Unitless	Liquid	Vapor	Liquid	Liquid	Liquid	Vap-Liq	Liquid	Liquid	
Temperature	°F	122	91	91	323	437	420	149	443	
Pressure	psia	354	45.0	45.0	45.2	45.2	45.2	1.1	1.3	
H ₂	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	
CO	lb/hr	87	87	1	trace	0	trace	trace	trace	

H ₂ O	lb/hr	122	48	72	2	trace	2	2	trace
CO ₂	lb/hr	767	737	29	trace	trace	trace	trace	trace
H ₂ S	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace
C ₁ -C ₄ (H ⁺)	lb/hr	2,948	1,990	940	17	0	17	17	trace
C ₁ -C ₄	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace
C ₅ -C ₇	lb/hr	7,589	466	4,617	2,505	2,239	4,744	4,744	trace
C ₈ -C ₁₆	lb/hr	7,119	trace	12	7,106	82,553	89,659	89,467	193
C ₁₇ -C ₂₂	lb/hr	116	trace	trace	116	3,711	3,826	930	2,896
C ₂₃ -C ₄₄	lb/hr	7	trace	trace	7	100	107	0	107
Total	lb/hr	18,754	3,329	5,672	9,753	88,603	98,356	95,160	3,196

Table 8: Mass Balance Hydrocracking Section-500 Part 1

Component	Units	F6	C1	C2	C3	T15	C4	C5	C6	C7
Phase	Unitless	Liquid	Liquid	Vap-Liq	Vap-Liq	Liquid	Liquid	Vap-Liq	Liquid	Liquid
Temperature	°F	3898	3898	3898	3898	0	0	3898	9179	9179
Pressure	psia	4456	4456	4456	4456	0	0	4456	9179	9179
H ₂	lb/hr	trace	trace	5,276	5,276	trace	trace	5,276	9,179	9,179
CO	lb/hr	5,120	5,120	5,120	5,120	trace	trace	5,137	trace	trace
H ₂ O	lb/hr	4,648	4,648	4,648	4,648	trace	trace	4,649	9,179	9,179
CO ₂	lb/hr	5,438	5,438	5,438	5,438	trace	trace	5,613	9,179	9,179
H ₂ S	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
C ₁ -C ₄ (H ⁺)	lb/hr	26,289	26,289	26,289	26,289	trace	trace	27,863	trace	trace
C ₁ -C ₄	lb/hr	trace	trace	18,124	18,124	trace	trace	19,398	1,254	1,254
C ₅ -C ₇	lb/hr	8,206	8,206	8,206	8,206	trace	trace	8,208	0	0
C ₈ -C ₁₆	lb/hr	3,828	3,828	3,828	3,828	0	0	3,828	0	0
C ₁₇ -C ₂₂	lb/hr	246	246	246	246	0	0	246	0	0
C ₂₃ -C ₄₀	lb/hr	38	38	38	38	0	0	38	0	0
C ₄₂ -C ₆₀	lb/hr	-27,590	-27,414	-26,117	-24,782	trace	trace	-25,364	trace	trace
Total	lb/hr	26,222	26,399	51,095	52,430	0	0	54,892	28,791	28,791

Table 9: Mass Balance Hydrocracking Section-500 Part 2

Component	Units	C8	C9	C10	C10B	C11	C12	C13	C14	C15
Phase	Unitless	Liquid	Vapor	Vapor	Vapor	Vapor	Liquid	Vapor	Liquid	Liquid
Temperature	°F	0	0	9179	0	0	9179	0	0	0
Pressure	psia	0	0	9179	0	0	9179	0	0	0

H ₂	lb/hr	trace	0	9,179	0	0	9,179	0	0	trace
CO	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
H ₂ O	lb/hr	0	trace	trace	trace	trace	9,179	0	0	trace
CO ₂	lb/hr	trace	0	trace	0	0	9,179	0	0	trace
H ₂ S	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
C ₁ -C ₄ (H ⁺)	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
C ₁ -C ₄	lb/hr	trace	0	trace	0	0	1,254	0	0	trace
C ₅ -C ₇	lb/hr	trace	0	trace	0	0	0	0	0	0
C ₈ -C ₁₆	lb/hr	trace	trace	trace	trace	trace	0	trace	0	0
C ₁₇ -C ₂₂	lb/hr	trace	trace	trace	trace	trace	0	trace	trace	0
C ₂₃ -C ₄₀	lb/hr	trace	trace	trace	trace	trace	0	trace	trace	0
C ₄₂ -C ₆₀	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
Total	lb/hr	0	0	9,179	0	0	28,791	0	0	0

Appendix V: Equipment List

The following tables highlight the main equipment needed to run the process as well as the recommended material of construction.

Table 10: Biomass Gasification Section-100

Equipment	Description	Material of Construction
DRY-100	Biomass Drier	Carbon Steel
R-100	Gasifier	Carbon Steel

Table 11: Syngas Cleanup Section-200

Equipment	Description	Material of Construction
CYC-200	Syngas Cyclone	Carbon Steel
V-200	Syngas Scrubber	Carbon Steel
V-201	Knock Out Drum	Carbon Steel
R-201	Water Gas Shift Reactor	1.25Cr-0.5Mo Alloy
V-202	CO ₂ Scrubber	Carbon Steel

Table 12: FT Reactor Section-300

Equipment	Description	Material of construction
R-300	FT Reactor	316 S.S
V-300	HP Flash Drum	Carbon Steel
V-301	LP Flash Drum	Carbon Steel

Table 13: Hydrotreating Section-400

Equipment	Description	Material of construction
R-400	Hydrotreater Reactor	1.25Cr-0.5Mo Alloy
V-400	Flash Drum	Carbon Steel
C-400	Distillation Column	Carbon Steel
C-401	Vacuum Distillation Column	Carbon Steel
P-400	Centrifugal Pump	Carbon Steel
P-400A	Centrifugal Pump	Carbon Steel
P-401A	Centrifugal Pump	Carbon Steel

Table 14: Hydrocracking Section

Equipment	Description	Material of construction
R-500	Hydrocracker Reactor	2.255Cr-1.0 Mo Alloy
V-500	3-phase Separator	Carbon Steel
C-400	Distillation Column	Carbon Steel
P-500	Centrifugal Pump	Carbon Steel
P-501	Centrifugal Pump	Carbon Steel
P-500A	Centrifugal Pump	Carbon Steel

Appendix VI: Equipment Sizing

The following table displays the equipment specifications in order to meet the requirements of our plant's production.

<i>Table 15: Heat Exchanger Duties</i>	
Equipment	Duty (KW)
E-100	5.33
E-101	13373
E-102	902
E-201	16976
E-300	9881
E-400	1864
E-401	2071
E-500	1341
E-501	4754

<i>Table 16: Pump Flow Requirements</i>	
Unit	GPM
P-500	449.64
P-501	13.55
P-500 A	65.17
P-400	65.85
P-400A	12.23
P-401A	26.54

Table 17: Column Sizing

Distillation column	Diameter (ft)	Height (ft)	R.R	D/F
C-400	3	58	0.395	0.48
C-401	19.5	70	0.080	0.967
C-500	9	64	0.669	0.237

Appendix VII: Sample Hand Calculations

Hand calculations were performed to validate the Aspen RADFRAC design for the C-401 vacuum distillation column by estimating (1) number of stages, (2) column height, and (3) column diameter. Because the system is multicomponent (SAF/diesel-range hydrocarbons), the Fenske–Underwood–Gilliland (FUG) method was applied: the Fenske equation establishes a lower bound on the required stages at total reflux, the Underwood equations determine the minimum reflux condition consistent with the specified split, and the Gilliland correlation relates reflux to the practical number of stages required under operating conditions. The hand calculations were intended to provide a close comparison to Aspen trends rather than replicate a full rigorous stage-by-stage solution.

The vacuum distillation column (C-401) was modelled with a total condenser. Aspen data extracted for the analysis included feed/distillate/bottoms molar flow rates and mole fractions, feed K -values (for computing relative volatilities), the reflux flow L and distillate flow D reported by RADFRAC, the feed thermal condition q , and tray-hydraulics quantities (internal vapor traffic and phase densities) from the Hydraulic Plots. Consistent with the Aspen split defined for

SAF and diesel-range material, the light key (LK) was defined as C16 and the heavy key (HK) as C18. The feed quality was taken as $q=0.96$, indicating a predominantly liquid feed entering the column.

The minimum number of theoretical stages was estimated using the Fenske equation, which provides the minimum stage requirement under the ideal limiting case of total reflux:

$$N_{\min} = \frac{\ln \left[\left(\frac{x_{LK}}{x_{HK}} \right)_{\text{dist}} / \left(\frac{x_{LK}}{x_{HK}} \right)_{\text{bot}} \right]}{\ln(a_{LK/HK})} \Rightarrow N_{\min} = 4.0 \text{ theoretical stages}$$

A small $N_{\min}$ indicates that the specified LK/HK cut is feasible from a volatility standpoint and does not require an exceptionally tall tower under ideal conditions.

To apply the multicomponent Underwood method, relative volatilities were computed using Aspen feed K -values referenced to the heavy key:

$$a_i = \frac{K_i}{K_{HK}}$$

Feed K -values were used as a representative basis because the ordering and relative behavior of K across the component slate remains consistent, even though absolute K -values vary by stage temperature and pressure.

The Underwood root equation was then solved for ϕ using the feed mole fractions z_i , the computed α_i , and $q=0.96$:

$$\sum_i \frac{q z_i}{a_i - \phi} = 1 \Rightarrow \phi = 1.9982$$

The solution was confirmed to fall within the expected bracket for an HK-referenced volatility set, $1 < \phi < \alpha_{LK}$. The minimum vapor rate was computed using the second Underwood equation:

$$V_{\min} = \sum_i \frac{a_i D x_{D,i}}{a_i - \phi} \Rightarrow V_{\min} = 661.7 \text{ lbmol/hr}$$

Considering that a total condenser is present, the minimum reflux flow was obtained from:

$$L_{\min} = V_{\min} - D \Rightarrow L_{\min} = 47.6 \text{ lbmol/hr}$$

with the distillate flow taken from Aspen as $D = 613.57 \text{ lbmol/hr}$, the corresponding minimum reflux ratio was:

$$\left(\frac{L}{D} \right)_{\min} = \frac{L_{\min}}{D} \Rightarrow \left(\frac{L}{D} \right)_{\min} = 0.0776$$

The operating reflux ratio was taken from Aspen RADFRAC results:

$$\frac{L}{D} = \frac{49.35}{613.57} \Rightarrow \frac{L}{D} = 0.0804$$

The operating reflux ratio being only slightly above $(L/D)_{\min}$ indicates operation near minimum reflux, which is consistent with requiring a larger number of stages than the minimum-stage limit.

The Gilliland correlation was then used to estimate the actual number of theoretical stages at the chosen operating reflux ratio. The Gilliland X parameter was computed as:

$$X = \frac{\left(\frac{L}{D}\right) - \left(\frac{L}{D}\right)_{\min}}{\left(\frac{L}{D}\right) + 1} \Rightarrow X = 0.00265$$

Using the correlation form for Y:

$$Y = \frac{1 - X^{0.0031}}{1 - 0.99357 X^{0.0031}} \Rightarrow Y = 0.7428$$

The stage requirement was then obtained from the standard Gilliland relationship:

$$Y = \frac{N - N_{\min}}{N + 1} \Rightarrow N = \frac{Y + N_{\min}}{1 - Y} \Rightarrow N = 18.48 \text{ stages}$$

To convert theoretical stages to actual trays, an overall tray efficiency of $E = 0.80$ was applied:

$$N_{act} = \frac{N}{E} \Rightarrow N_{act} = 23.10 \approx 23 \text{ trays}$$

This tray count is consistent with a practical vacuum fractionation cut and reflects the expected increase in required stages when operating close to minimum reflux.

Column height was estimated using a standard tray spacing of 2 ft and additional allowances for top and bottom equipment/disengagement regions:

$$H = (N_{act} - 1)(2 \text{ ft}) + 4 \text{ ft (top)} + 8 \text{ ft (bottom)} \Rightarrow H = 56 \text{ ft}$$

Finally, column diameter was estimated by tray hydraulics using the maximum internal vapor-traffic stage identified in Aspen Hydraulic Plots (Stage 21). Flooding velocity was estimated using the Souders–Brown form with tray capacity constant C , and a design velocity at 80% of flooding was applied:

$$u_f = C \sqrt{\frac{\rho_L - \rho_V}{\rho_V}} \Rightarrow u_f = 3.34 \text{ m/s}$$

$$u_{des} = 0.8 u_f \Rightarrow u_{des} = 2.67 \text{ m/s}$$

Diameter was then computed using the vapor mass flow rate $\dot{m}_V$, vapor density ρ_V , and an active area fraction $\phi=0.8$ to account for tray downcomers and non-active cross-sectional area. A tray capacity constant of $C=0.11 \text{ m/s}$ was used:

$$D = \sqrt{\frac{4 \dot{m}_V}{\pi \rho_V \phi u_{des}}} \quad (\phi=0.8, C=0.11 \text{ m/s}) \Rightarrow D = 4.16 \text{ m} = 13.6 \text{ ft}$$

The resulting diameter reflects the large cross-sectional area typically required for vacuum towers, where low vapor density increases volumetric flow and hydraulics become controlling. Differences between the hand diameter and Aspen-reported diameter are expected because Aspen tray sizing incorporates additional constraints (e.g., downcomer backup, entrainment, weeping, and tray geometry assumptions) and may be governed by a different internal location depending on the controlling hydraulic limit.

For comparison to Aspen's reported stage count, RADFRAC reports theoretical stages only (excluding condenser and reboiler). Aspen reported 22 theoretical stages for C-401; applying the same overall tray efficiency assumption ($E=0.80$) gives an estimated physical tray count of $22/0.80 \approx 27$ trays, which is consistent in form with the hand-correction procedure used above. The Aspen results and hand calculations are compared in the table below

<i>Table 18: Hand vs Aspen Calculation for Column C-401</i>		
C-401	Hand Calculation	Aspen calculation
Column Trays	23 trays	27 trays

Column Diameter	13.6 ft	14.0 ft
Column Height	56 ft	70 ft

As shown in *Table 18*, although Aspen predicts a larger column height than the hand estimate, the simulated sizing results remain consistent with typical industrial vacuum column dimensions and provide a reliable basis for preliminary sizing.

Appendix VIII: Preliminary Utilities

The following, *Table 19*, displays the preliminary utilities needed to operate the process. These specifications were obtained from the process simulated on ASPEN v14.

<i>Table 19: Cooling Water Flowrates</i>	
Equipment	Flowrate (gpm)
E-101	68046
E-102	3339
E-201	64182
E-300	37355
E-401	7829
E-400A	23196
E-401A	2209
E-501	17975
E-500A	9589

Total	233720
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<i>Table 20: Electricity Requirement for Equipment</i>	
Equipment	Power (kw)
K-100	5.33
K-200	13204
K-201	11310
P-400	26.72
P-400A	0.023
P-401A	0.049
K-400	14.31
P-500	297
P-501	11
P-500A	0.17
K-500	2180
Total	27048.6

<i>Table 21: Steam Requirements for Equipment</i>			
Equipment	Duty (kw)	Flow (lbs/hr)	Type
E-100	0.979	3.54	LP
E-400B	1011.3	3944.6	MP
DRY-100	73751.6	340447.7	HP
E-500	1337.2	6172.6	HP
E-500B	7621.0	35179.8	HP
E-401B	1540.5	7111.1	HP

<i>Table 22: Total Steam Consumption by Type</i>		
Type	Duty (kw)	Flow (lbs/hr)
LP steam	0.979	3.54
MP steam	1011.3	3.944.6
HP steam	84250.3	392859.34

<i>Table 23: Natural Gas Consumption by Equipment</i>		
Equipment	Duty (kw)	Flow (lbs/hr)
R-100	21736.06	107819.5

E-400	1857.551	9214.197
R-101	34166.13	169477.7
Total	57759.74	286511.397

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